

6.11 SOAP AND SYNTHETIC DETERGENTS

Introduction

The use of soap or some natural soap-like cleaning agents has always been associated with man's inherent instinct to keep his body, clothes and washable belongings clean.

Soaps and synthetic detergents, although they differ markedly in their chemical structure, are grouped under a common class of surface active agents (surfactants) because their aqueous solutions have many surface active properties in common, e.g. lowering of surface and interfacial tension, wetting, foaming, emulsifying, etc. The magnitude of such properties is governed by factors like structure of surfactant, concentration and temperature of solution, and the presence of certain additives.

Soaps suffer from certain drawbacks which have caused their replacement to an increasing degree by synthetic detergents both in household and industrial applications.

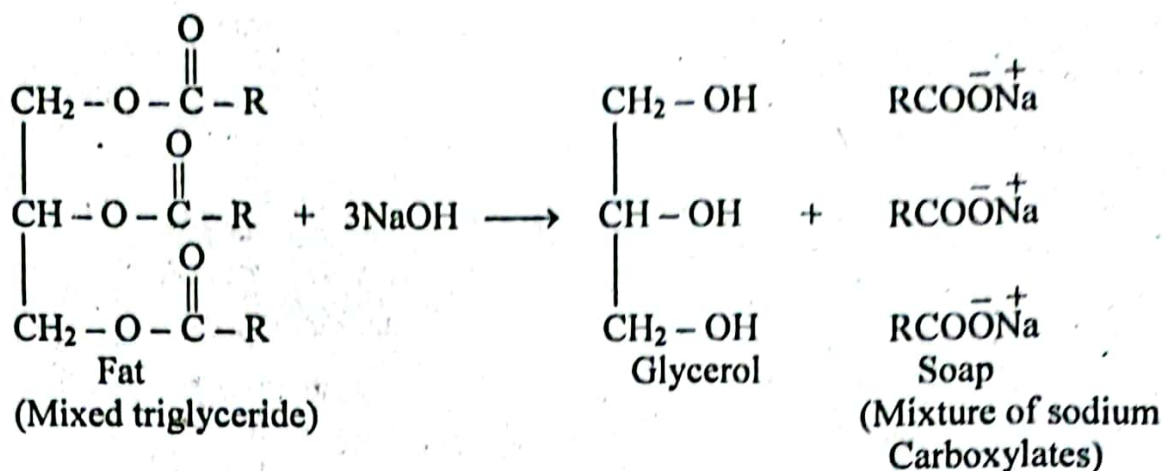
First, they do not wash well in hard water because of the precipitation of insoluble calcium and magnesium soaps in the form of a scum which sticks to fabrics that are being washed and promotes adhesion of dirt. This is the reason for the grayish colour which results when white cloth are repeatedly washed with soap in hard water.

Secondly, soaps are not stable even under mildly acidic condition (in fact below pH of 9) on account of their hydrolysis and decomposition into fatty acids and acid soaps. They are, therefore, ineffective in many textile processing operations.

Thirdly, they have a limited storage property so that they deteriorate with time. These drawbacks led at first to the development of new surface active agents for industrial applications. Later, and especially over the last 40 years, these detergents began to replace soap in many household applications to achieve ultimately their present tremendous importance.

Soap

Soap is defined as ordinary soaps are the products of hydrolysis of oils and fats with sodium hydroxide, as alkali salts of fatty acids with a restricted chain length, mostly C_{12} to C_{18} . The commercial soaps are mixtures of several such salts and some additives intended for supplementing the detergency as well as to make them more attractive in respect of their odour and colour.



In addition to their use as cleaning agent, soaps play a key role in the emulsion polymerization processes used in the rubber and plastics industries. In the textile industry, soaps find use in kier-boiling cotton, scouring wool and degumming silk to remove impurities prior to any processing operation. They are widely used in the cosmetics industry in the preparation of a variety of skin cleaning and conditioning emulsions. Sodium and lithium soaps are extensively used to thicken mineral oils in the manufacture of lubricating greases.

Leather is decreased by a hot soap solution during processing. Soaps are still used to some extent as wetting and spreading agents to improve the dispersal of insecticidal and fungicidal components of agricultural sprays. Soaps also find use in pharmaceuticals, and in road building and maintenance.

The processes and plants employed by the soap industry have come to be developed by empirical methods and experience. It was only in the present century that science entered soap-making in a big way, giving a clearer picture of the reaction taking place in the soap kettle and of the properties of soap in solid, liquid and solution forms.

The advances in different branches of engineering have contributed to a rapid development in soap technology. Use of non-rusting and non-corrosive alloys in plant construction has led to improvements in quality of products.

New techniques, continuous operations, novel designs and automation have resulted in a better efficiency and economy. High pressure saponification, centrifugal separation of soap, counter current washing, chilling on rolls, continuous drying, dehydration by spraying and vacuum plodding may be cited as examples of novel applications of equipment and operations.

In addition, use of new raw materials and new methods of processing conventional raw materials have widened the field of selection and made the industry more flexible.

Raw Materials

Alkalies and fatty materials are the basic raw materials for soap. Many other ingredients are also added to maintain its good detergency as well as to increase its saleability.

The important raw materials are (i) alkalies, (ii) fatty materials, (iii) rosin and tall oil (iv) sodium chloride, (v) builders, (vi) fillers, and (viii) auxiliary materials.

Alkalies

In soap-making, alkalies are used to saponify fats and oils. Caustic soda is the most important alkali in the manufacture of hard soaps. Caustic potash is used for the preparation of soft soaps (liquid or paste). Blends of the two alkalies are occasionally used in order to achieve special properties.

Ammonium hydroxide and sodium carbonate, being non-saponifying alkalies, are used only with fatty acids. Ethanolamines combine with fatty acids to give highly soluble soaps.

Fatty materials

Any fat or oils can be used for soap-making. Some of the important fats and oils used in soap industry are classified into three types, e.g., (i) nut oils, such as coconut, palm kernel; (ii) soft oils such as groundnut, cotton seed, olive, castor, linseed, greases and (iii) hard fats, such as animal fats (tallows), palm, mahua, hardened fish oils, hardened vegetable oils (vegetable tallows).

Nut oils produce soaps which lather profusely. Soft oils give soaps which have the desired ready solubility in water.

Hard fats produce soaps having a firm texture and hardness. Many other oils and fats peculiar to particular regions are used. A judicious selection from these types is made according to the availability and prices of the fats and the desired properties of the end products.

Fatty acids obtained by splitting of oils and fats may be used partially or wholly in place of oils if economics permits for the preparation of special soaps. Soapstock obtained as a by-product in the refining of oils and fats serves also as a cheap raw material in soap-making. It contains soap, neutral oil and water. Soapstock from a soft oil like groundnut oil comprises 40 – 50 percent of neutral oil.

Rosin and tall oil

Both of these are obtained from pine trees. In the form of oleo-resin (solution of resin in turpentine) oozes out as a sap when incisions are made in the bark of living pine trees. The oleo-resin is steam distilled to yield rosin as a residue.

Rosin is a non-fatty acidic substance whose sodium salt has desirable properties like increased solubility and lathering in hard water. Rosin polymerized or hydrogenated to different degrees is also used in soap-making.

Tall oil is a waste product obtained in the manufacture of paper pulp from pine wood. It consists of 40 – 60 percent rosin acids, 40 – 60 percent fatty acids and 6 – 10 percent sterols and higher alcohols.

Sodium Chloride

It is an essential raw material in the graining of soap in the full boiled process and may be used either in powder form or as brine. The salt used should be free of any iron, calcium or magnesium compounds, as they cause deterioration of soap.

Builders

These are special materials added to soap to maintain or improve its effectiveness under different conditions of use. Some of their specific functions are.

- (i) To maintain the proper pH of soap.
- (ii) To prevent precipitation of soaps in hard water.
- (iii) To improve the soap's texture.
- (iv) To effect economy in soap usage without effecting detergency.

Some important builders are sodium silicates, sodium phosphates, sodium salt of ethylenediamine tetra-acetic acid (EDTA), soda ash, borax, starch and a few cellulose derivatives.

Fillers

These are mostly inert substances. When added to soap, they increase its bulk/weight but reduce its effectiveness. Being much cheaper than soap, they are incorporated in it to give a low priced product. Common fillers are clays, talc, starches, sodium sulphate, chalk and magnesium carbonate. Soda ash and silicates are also fillers when used in large quantities.

Other auxiliary materials

Antioxidants are added to soap to prevent it from rancidity. Sodium silicate, sodium hyposulphite, sodium thiosulphate, stannous chloride and EDTA are some of the important antioxidants. Colours and perfumes are also added to soaps to enhance their appeal to the consumer.

Pre-treatment of Oils

Fats and oils in their crude conditions vary in their qualities, such as appearance, smell, colour, presence of impurities etc. They, therefore, need some pre-treatment before saponification so that soaps of good, uniform quality are obtained.

Kind of pre-treatment and its extent depend upon the quantity of the available raw oil and the desired quantity of the end product as also the economics of the treatment. These are matters of judgment on the part of the soapmaker.

Degumming

Suspended and colloidal impurities (called gums) are normally removed by a hot brine or steam treatment. The gum may float on top or may precipitate. Their separation is facilitated by use of a centrifuge.

Alkali Treatment

Oils which contain excess of free fatty acids are given alkali treatment. Aqueous sodium hydroxide equivalent to the free acid content of the oil is added with slow stirring to the hot oil. Crude soap is precipitated and it absorbs some of the unwanted colour bodies. It is then allowed to settle for some time after which the neutral oil is filtered off.

Bleaching

This treatment is given to decolourize the oils. Materials like activated Fuller's earth or activated carbon have the property of removing colour from many liquids by adsorption. A measured quantity of the earth or the carbon or both is mixed with the oil, agitated for some time and the mixture is filtered when a pale oil is obtained. The quantities of the absorbent (0.5 – 2.0%), temperature of bleaching (60 – 90°C), time of contact (20 – 40 min), type of agitation, etc. have to be fixed by trials. Bleaching is more effective under vacuum.

There are other methods—some mild and some drastic at about 100°C wherein oxidizing chemicals are used. Hydrogen peroxide, organic peroxides, sodium hypochlorite, sodium chlorite and sodium dichromate are such chemicals used at 0.5 – 2.0 percent level. Palm oil (deep orange colour) is bleached by bubbling air through it at about 100°C.

Hydrogenation

Some highly unsaturated oils like fish oils and linseed oil are hydrogenated to a semi-hard or hard condition and used as a hard stock. In many cases partial hydrogenation improves the stability, colour and odour of the oil.

MANUFACTURE OF SOAP

Soap making involves broadly two distinct operations, namely, saponification and finishing, both of which may be carried out either by a batch or a continuous process.

Batch process

There are three conventional batch processes of saponification, e.g., (i) cold process (ii) semi-boiled process and (iii) full boiled process.

Cold process

This is the simplest method of saponification and requires minimum of equipment. In it, advantage is taken of the natural heat generated during the reaction of the oil and alkalies. The mixture of oils is taken in a convenient vessel (a cylindrical or rectangular and flat or dish bottomed pan). If the charge contains tallow or rosin, the pan

is warmed till the whole mass is in a liquid condition. Calculated quantity of caustic soda solution is very gradually added to the oil mixture keeping the mass continuously stirred.

As the mixing proceeds, the mass thickness and stirring is continued till the soap is smooth, homogenous and firm enough. Filters, perfumes and colour are added at this stage. This thick mass is then poured into frames where saponification is completed during cooling and solidification. A setting time of 2 – 3 days is given to the soap in frames. The solidified mass have the same size and shape of the frame is taken out, cut into slabs and then into bars and stamped. Mostly hand-worked machines are employed for these operations.

Semi-boiled process

The equipment used in this process consists of a pan which is fixed over a stationary fireplace suitably designed for controlled and uniform heating and for removal of hot flue gasses. The general procedure is to melt the fatty stock in the pan along with some soap scrap and to raise the temperature of the mixture to about 80°C. The alkali solution (as such or mixed with sodium silicate) is run in and temperature is so maintained as to keep the mass boiling. The high temperature permits speeding up of saponification as well as its completion.

The mass is kept alkaline to prevent bunching. During boiling, the required fillers and builders may be added under vigorous agitation by a hand-operated pole stirrer. Care is taken not to allow the pan to boil over. Towards the end, when the mass is smooth and homogenous, perfume and colour are added. The whole mixture is dropped into frames and allowed to cool. The cooled mass from the frames is then finished into bars, cakes or flakes as desired.

This process is popular with small soap-makers because of its simplicity, and is used for making household soaps, soft soaps (mostly for textile industry) and shaving sticks.

In both the cold and semi-boiled processes, glycerine which is liberated during saponification remains with the soap and is thus lost.

Full boiled process

This is widely used by large soap-makers all over the world. The boiling of soaps is carried in kettles housed in a multistoried building. The kettle which is well lagged is provided with a closed steam coil and one more open coil at the bottom to allow steam into the soap mass. A skimmer pipe is used to remove the neat soap from the top without disturbing the bottom nigre layer after the fitting and settling operations. The pipe is held vertically during boiling and lowered when the neat soap is ready for removal. Inlets for caustic soda, oil, water and brine are provided at the top of the kettle. Often the kettle is covered and provided with ducts to remove the escaping steam with an exhaust fan. In this case, the covers are provided with lighting arrangements and sight glasses to watch the progress of the boiling operations. Since it takes about a week to boil a batch and

draw out the neat soap, there are a number of kettles in operation. So that the neat soap is available from one kettle in rotation for further processing everyday. The kettles are mostly of mild steel, but may be of monel, nickel or stainless steel when good quality toilet soaps free from iron contamination are to be manufactured. Their capacities may vary from a few tones to 100 tonnes per batch.

The auxiliary equipment for kettles consists of several tanks on the ground level for the preparation of alkali and brine solution and for storing half spent lye, spent lye and nigre and overhead tanks for carrying these various solutions and water. All these tanks and kettles are connected by pumps, pipes and valves.

Stages in soap-boiling operations

The various stages in the soap-boiling operation are called changes. These are

(i) saponification (ii) brine change (iii) strong change (iv) rosin change (v) fitting

Rosin change and fitting are carried out only once, while brine change and strong change may be carried out a number of times. The actual number of these is governed by two main consideration, i.e; (i) quality of the product and; (ii) glycerine recovery. Better colour of the neat soap and maximum recovery of glycerine are possible only with a number of brine changes carried out in a counter-current manner.

Saponification

There are great variation in starting soap boiling. One technique is to start a fresh boil on the nigre (deep coloured impure soap) left over from a previous boil. The nigre is boiled up by open steam. Portions of oil charge, half spent lye and fresh lye are added in portions to the boiling mass to bring utilize the free caustic alkali present in the nigre and half spent lye.

At first, the oil gets emulsified and the saponification starts. The addition of soda and oils is contained in instalments as saponification proceeds briskly. As, in the absence of sufficient alkali thickening or bunching of the soap (lump formation) may take place, it is always advisable to charge initially into the kettle some quantity of solid salt or salt solution, sufficient water is added periodically to keep the mass thin. Boiling is continued till practically the entire quantity of oil is used up and a smooth homogeneous soap is produced. Before graining, the soap mass is boiled with further quantity of oil till no free caustic soda is present. Complete saponification is effected during the subsequent changes.

Brine change

The object of the step is to separate glycerine in the form of an aqueous solution (called lye) from the soap. On treatment with salt, the soap, grains, out on top while the graining form the lower layer, when the hot mass is allowed to settle in the kettle.

This operation is, therefore, called graining. The soap is kept boiling gently and salt (fine solid or brine) is added uniformly over the surface in portions.

On adding sufficient salt, the mass becomes heterogenous. When a sample, at this stage, is taken out on a trowel, the viscous grained soap and the lye separate in two phases and trickle down when the trowel is held at a sloping angle. The kettle is then allowed to rest for 3 – 5 hours, preferably overnight, for separation of the grained soap and the spent lye. The lye is withdrawn and sent for recovery of glycerine. More than one brine change may be given to improve the colour of soap and to increase the amount of glycerine recovered.

Rosin Change

Rosin is not added to the kettle during the first saponification change, but separately saponified in a small kettle using soda ash or caustic soda. Salt is added to grain out the soap and the lye containing the impurities is discarded since it does not contain any glycerine. Grained rosin soap settles to the bottom unlike the fatty acid soap. This rosin soap is then added to the grained soap mass during the strong change described below.

Alternatively, the required amount of rosin may be added to the grained soap in small lumps and boiling continued with further addition of caustic soda. The rosin melts and is saponified.

The kettle is tested for absence of free caustic alkali (tongue test). It is then grained with further addition of salt, if necessary, and allowed to settle after which the lye is withdrawn.

Strong Change

The object of the strong change is to boil the mass with excess of alkali so that complete saponification takes place and no free fat is left over. The grained soap from the rosin change or the brine change, as the case may be, is boiled and water is added to bring it to a homogenous condition. Caustic soda is added and boiling is continued with open steam till the caustic soda is taken up by any unsaponified oil and a grained with strong caustic soda solution or the second half spent lye from a previous kettle, if the counter current system is adopted.

A second strong change may be given in a similar way. The lye separated here (half spent lye) is utilized in the first strong change.

Fitting

This change (called pitching) causes the soap to separate into two phases, neat soap (pale coloured purified soap) and nigre (deep coloured impure soap), and is the most critical operation in soap making because the final qualities of soap i.e., colour, texture, alkalinity, salt content etc depends upon it. In actual fitting operation the kettle after withdrawing the half spent lye from the previous caustic change is boiled and the condition of the soap is noted when a sample is taken on a trowel. If large soap grains run off the tilted trowel, water is to be added, if the soap sticks to the trowel to form a heavy coat, small quantities of alkalies/salt are to be added. A 'fit' is indicated when the soap

slides off the surface of the trowel in the form of transparent flakes and the surface of trowel becomes dry immediately.

When the proper degree of fit is obtained, the kettle is allowed to settle. The time of setting depends upon the size of the kettle, amount of soap and the type of fit, and may vary from 1 to 2 days. After proper settling, the neat soap on top is withdrawn through the skimmer pipe and pumped to storage tanks kept hot by means of steam coils on the outside of the tanks. The nigre at the bottom is withdrawn.

The economics of soap boiling by the full boiled method is governed by two important factors – recovery of glycerine and proportion of nigre to neat soap. The first factor is dependent upon the number of brine changes, the amount of lye produced per unit of soap and the concentration of glycerine in the lye. The economics is improved by counter – current washing of soap during brine change. The influence of the second factor is obvious. The nigre has an average fatty acid content of 35 percent while the neat soap compositions is fairly constant, lying between 61 and 63 percent total fatty acids. Too large a quantity of nigre reduces the yield of neat soap, too small a quantity of it tends to impair the quality of soap.

Continuous Processes

Several continuous processes for soap manufacture have been developed in order to shorten the processing time and generally improve upon the conventional kettle method described above which is, no doubt time consuming.

High speed dispersion of the oil and its multiple contacting with alkali so as to accelerate the rate of saponification, use of centrifuges for quick separation of the phases, control of temperature, pH, alkali and salt contents at various stages, and use of non-corrosive materials of construction are the basic features of these processes and they have led to a tremendous saving in time, increase in efficiency and upgrading the quality of soaps.

The soap finishing process too has undergone remarkable changes, thereby eliminating certain conventional equipment. Some of the representative processes are described below.

Sharpless process

The sharpless process comprises the following four stages carried out in that sequence without any break.

(i) Partial saponification (ii) completion of saponification (iii) soap washing (iv) fitting to neat soap.

Neat soap is produced in less than two hours. The operating temperature and pressure do not exceed 105°C and 3 kg / cm^2 respectively. The main feature of the process is the use of centrifuge at all the four stages for separating the phases.

Each centrifugal separation is preceded by mixing of the reagents and the intermediate product in a special mixer. Feeds at every stage controlled by means of flow meters and proportioning pumps.

Mon Savon Process

In this process, an emulsion of the fat in caustic solution prepared by passing them through a colloid mill is heated to 100°C. Saponification takes place immediately.

The soap flows down a column across baffles to a tank which acts as a buffer reservoir in which saponification is completed. The crude soap from here is pumped up through a multistage washing tower in a counter-current manner against a descending steam of hot brine solution. The soap flowing out at the top is fitted with caustic soda solution in another pan while the spent lye is continuously withdrawn from the bottom.

De laval Centripure Process

In this process, saponification, washing and fitting are carried out in succession in several columns joined in series. Saponification is accelerated automatically by recycling a portion of the soap already formed. Phase separation is done by centrifuges. Every stage is under automatic control. A noteworthy feature of the process is that the feed of fat and lye and the contents of alkali and salt are controlled by viscosity-sensing devices.

Crossfield process

The plant required for saponification by the Crossfield process consists of a shallow rectangular pan with three interconnected compartments. Metered quantities of fat and caustic soda solution are first emulsified to enhance saponification and the emulsion is then led into the shallow pan and is saponified as it passes from one compartment to the other in a quiescent flow.

Saponification control is maintained automatically by regulating the feed of reactants by means of a viscosity indicator in the recirculating loop. A bleed from the last compartment feeds a liquid-liquid extraction unit for counter-current washing with hot brine.

Fitting is done in a shallow tank having a pump-operated recirculation loop. Another viscosity control adjusts the electrolyte concentration for neat-nigre separation in a bank of centrifuges.

Meccaniche modern process:-

Saponification is carried out in this process under pressure and the subsequent stages of washing and fitting are done counter-currently in towers provided with separations.

Soap from fatty acids

With the establishment of high pressure, high temperature continuous fat-splitting and high vacuum fatty acid distillation cuts of fatty acids having the desired

properties are now easily available. These are neutralized with caustic soda to give a soap which may be processed further. Processes of Mills, Armour and Mazzoni are illustrative of the use of fatty acids.

In one Mazzoni process, fatty acids and caustic soda solution pre-heated to about 60°C are propelled by duplex pumps in to a multistage soap stream. Saponification temperature is maintained at 90°C. An amount of soap equal to the feed volumes is constantly withdrawn from this system to holding tank. Sodium chloride is either added with the caustic soda or separately.

In a variation of the Mazzoni process, sodium carbonate is used for most of the fatty acid neutralization and sodium hydroxide is used only for final adjustment.

Finishing of the Soap

The neat soap which is the end product of both the kettle and continuous saponifications is further processed for making marketable products, either washing or toilet soaps. There are, thus, two streams which neat soap follows.

In the traditional technique, the sequence for washing soaps was crutching, frame or press cooling, slabbing, cutting, air drying, stamping and packing.

The sequence for toilet soap was chilling, flaking, chamber-drying, mixing, milling, plodding, cutting, stamping and packing. Note all these operations are continuous.

There has now been a great revolution in the finishing operations which has been brought about mainly by the introduction of spray drying of the neat soap (this has eliminated frame and press drying and also roll chilling) and of powerful duplex vacuum plodders (this has eliminated milling). An additional economic advantage has been the considerable reduction in soap scrap.

Washing Soaps

In the conventional drying method, the molten neat soap is fed into a crutcher (a jacketed vessel provided with a powerful agitator) to which are added colour, perfume and other ingredients under agitation. The hot mass is then fed to a press-cooler (frames cooled by chilled water) and allowed to remain there for about 2 hours. The frames are then opened whereby slabs of soap are obtained. These are cut into bars which are stacked for air drying in the open. The moisture content is controlled at about 30 percent in the crutcher itself.

In a modern Mazzoni plant, vacuum spray drying technique is employed. The hot laundry soap base mixed with the additives in the crutcher is sprayed through a nozzle into a cylindrical chamber kept under a pressure of 27 – 28 mm of Hg.

A part of the moisture from the soap mix evaporates off which, thus, cools the sprayed mass. The cooled material in the form of powder adheres to the sides of the chamber from where it is scraped off and withdrawn from the bottom and led into a plodder?

plodding's cutting and drying

The scraped material (laundry soap) from the vacuum spray dryer is fed into a set of two vacuum plodders arranged in series. An internal screw in the plodder forces the soap into a compression area terminated by a tapered outlet which is fitted with a perforated disc in the first plodder. The soap comes out in the form of nodules which are fed into the other plodder in which the disc at the outlet is replaced by a die through which a 'log' of soap is continuously extruded. This is automatically cut into required bar lengths which are then conveyed over a band through an air heated enclosure. This achieves surface or skin drying to enable the bar to be stamped easily. This is the modern plodded soap bar in contrast to the earlier cut bar.

The mechanical treatment under the duplex vacuum plodder enriches the soap with β -phase (almost 100%) and endows it with good leathering property.

Toilet Soaps

In the traditional drying method, the molten soap is fed on to a rotating chilled roll which is provided with a spreading roll. The spread mass is scraped off to form ribbons which are converted to endless band. This enters a heated cabinet dryer in which conditions are so adjusted that the emerging ribbons have a moisture content of about 15 percent.

The flash spray drying of toilet soaps has been introduced in modern plants. The hot neat soap is superheated to a temperature of about 150°C and is sprayed through a nozzle at the top into cylindrical vessel which is under atmospheric pressure and is provided with a continuous stream of hot air from below.

The conditions are so adjusted that the flashed steam is removed immediately and the moisture content of soap is reduced to 12-15 percent. The mass adhering to the inside of the vessel is scraped off and extruded from the bottom in the form of chips or flakes and stored into bins.

Mixing, milling and plodding

Pigments, dyes, perfumes and other chemicals are generally added to the dried toilet soap ribbons, chips or flakes in a batch mixer (amalgamator) equipped with a helical agitator. The mixture is then dumped into a three or five roll mill, where it is squeezed through the first two rolls, picked up by the faster roll and passed onto the final roll from where the mass is scraped off in the form of ribbons. These are fed to a simple plodder fitted with a die at the tapered end. The emerging log of soap is automatically cut to the size of cakes, stamped and packed.

Soap Products

Soaps come in three physical forms, i.e., (i) solid (ii) disintegrated and (iii) non-solid.

Solid

The soap base, being plastic, is able to take up any desired shape under pressure and retain it. There are many common solid forms like bars, tablets, sticks and cakes which can again be round, spheroidal, elliptical, oval, or of any kind of fancy shape.

Disintegrated

The soap base after drying is also made available either in a coarsely disintegrated form like chips, shavings and nodules or in a still finer form like beads, grains, flakes and powder. These are the preferred forms of washing soaps, when clothes are not to be scrubbed. Ease of solubility and convenience in use of this form of soap result in saving of labour and time.

Non-solid

These are paste, jelly, cream and liquid in which form the products are easily soluble. Their preparation involves some alteration in the technique of manufacture.

Soaps are also classified into categories like toilet soap, household soap, medicated soap and industrial soap according to the various uses to which they are put.

Toilet Soaps

The toilet soap in any one of its forms has to be of high quality and mild on skin as it is used for personal cleaning. It also has to have consumer appeal in its shape, size, colour and fragrance. The special operation of mixing, milling and plodding produce a glossy, smooth and attractive cake.

Transparent Soap

Transparent soap is a type of toilet soap so called because of its transparency (correctly speaking, the soap is translucent and not transparent). It is, however, not superior to the genuine, milled toilet soap in cleansing. It is the polished surface, lustrous, coloured appearance and translucency that catch the fancy of the consumers.

The soap is manufactured from full-boiled, fitted and dried soaps. Alcohol, sugar solution and glycerol are added to the hot soap to give it a glossy, translucent finish.

Shampoos, Shaving Soaps and Shaving Creams

Shampoos, shaving soaps and shaving creams (both lather and brushless varieties) are some of the other soaps used purely for toilet purposes.

The function of a shampoo is to cleanse the hair and leave it soft, lustrous and manageable after the soap has been washed away.

Originally, shampoos were based on soaps only. Incorporation of synthetic detergents improves the wettability of hair, lathering and rinsability of shampoos considerably.

Shaving soap, on account of its specific application, must have certain essential properties like quick wetting and profuse and heavy lathering. It should be mild on the skin, be a little softer than the usual toilet soap and yet should not wear away too soon.

These properties are obtained by the use of caustic potash partly in place of caustic soda and a large proportion of high titre fat and stearic acid. The oil base of shaving soaps is selected from coconut oil, palm kernel oil, tallow, highly hardened fats, palm oil and stearic acid.

Small quantities of glycerine, gum tragocantb or gum arabic are also added sometimes. Shaving creams are generally prepared by saponifying a mixture of cocnut oil and stearic acid with mixed caustic potash and soda. These are usually made by the semi-boiled method. Shaving bars and sticks are of similar composition, but for the water content.

Household Soaps

These comprise various types of laundry soaps available in the foam of bars, cakes, chips, flakes, powders, etc. Genuine soaps in the foam of bars contain about 30 percent moisture and not less than 67 percent anhydrous soap. Powder may not contain more than about 2 percent moisture. The anhydrous soap content of built soaps will be around 50 percent while that of filled soaps between 30 and 40 percent.

Two main types of laundry soaps are white and yellow. White laundry soap is usually a full boiled product made from 25 – 40 percent coconut oil and 60 – 75 percent tallow, hydrogenated fish oil, etc. It is always prepared with sodium silicate or other soap builders in sufficient amount to reduce the anhydrous soap content to about 50 percent. Such soaps are never milled.

Yellow laundry soap is similar to white laundry soap except that it contains a substantial proportion of rosin. The rosin imparts a distinctive yellow colour to the bar. Rosin soap is relatively soft and sticky and hence, yellow soaps cannot be prepared with as much builder as white soaps. Its anhydrous soap content is usually over 50 percent. It is also usual to employ less coconut oil in yellow than in white laundry soap because of the greater solubility of rosin soaps.

Household chipped or flaked soaps are generally a little lower in moisture content than the corresponding bar soaps. A representative composition is moisture 10 – 20; soap, 50-60; tetrasodium pyrophosphate, 15; sodium metasilicate, 7; and soda ash 8 percent.

Medicated soaps

Every soap has some hygienic effect. The soaps of fatty acids having a chain length of 8 – 10 carbons are effective against many pathogenic organisms. Thus, ordinary soap and water may be adequate against mild bacterial skin infection. However, to cope with more serious conditions of contamination, the medicinal properties of soap have to be enhanced. Such soaps are called medicated soaps.

Medicated soaps may be of different types, e.g., germicidal, bactericidal, disinfectant, sanitary, therapeutic, veterinary etc; depending upon the functions they have to fulfil. The ingredients generally added to toilet soaps to enhance their medical properties phenol creosote oil, formaldehyde, essential oils, pyrethrum, salicylic acid,

iodine and iodides, mercury salts and various synthesized chlorinated phenolic compounds.

Some soap preparation, e.g., *sapo mollis* (soft soap) and *liquor cresolis saponatus* (cresol and soap solution) are recognized medicinal preparations and are included in the British Pharmacopoeia (1968).

The former is a 50 percent soap product used in the preparation of liniments. The latter is the product of the saponification of linseed oil with caustic potash in the presence of cresol. Its final water diluted form contains 50 percent cresol and 20 percent soap, and it is used as an antiseptic and disinfectant.

Insecticidal soap used in agricultural sprays for controlling insect pests is also a form of medicated soap, although the action of soap in such insecticides is limited more or less to facilitating the contact of the actual insecticides with the insects. Lowering of surface tension and quick spreading are the beneficial properties soap imparts to such sprays.

Industrial Soaps

Textile soaps, soft soaps, liquid soaps, mechanic's soap and dry cleaner's soap are some of the important industrial soaps.

Textile soaps

Soda soaps of low titre are widely used in textile industry for scouring raw wool to remove wool grease, scouring the yarns and cloth after oiling and for fulling or milling. These soaps are also used for degumming raw silk and for cleaning rayon, silk and cotton, before and after dyeing and printing.

Soft soaps

These are used mainly as wetting agents and for cleaning decks of ships. Usually made by saponifying semi-drying and drying oils by the semi-boiled method, soft soaps have a translucent appearance and are pasty in consistency. Their soap content is about 40 percent.

Liquid Soaps

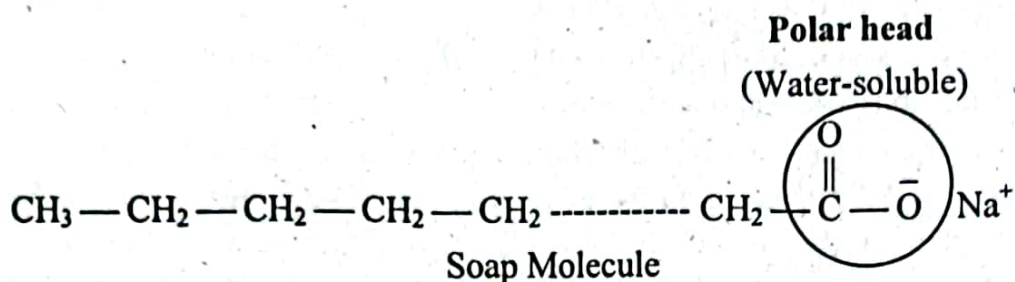
These find wide use as industrial toilet soaps for hand washing in factories, institutions, offices, hospitals and public wash rooms. The main feature of liquid soaps is the convenience with which they can be used through the dispensers. Liquid soaps are also used for scrubbing floors and as medicated or sanitary soaps after introducing appropriate ingredients in them.

Liquid soaps are manufactured by the semi-boiled process. Some of the desired properties of liquid soaps are clarity, good lather and detergency, mild action, pleasing aroma and suitable viscosity. Liquid soaps are based on potash saponified coconut oil but may contain in addition, soaps of rosin, oleic acid and soft oils like soyabean, castor and linseed. To increase clarity, sugar, glycerine or alcohol may be added.

Mechanic's Soaps

These are similar to scrubbing powders with high abrasive percentage. The function of the abrasive (ground silica or pumice) is to dislodge the grime sticking to the hands of the mechanics. The abrasive (25 – 30%) and other ingredients like soda ash and borax are crutched into the semi-boiled soap and the mixture is framed.

Cleansing Action of Soap



Skin is covered with a thin layer of natural oil or grease which catches dirt. The soap molecules have a long-chain non-polar hydrocarbon tail that is oil-soluble, and a negative head (COO^-) which is water-soluble. As soap water is poured over the skin (or a dirty garment), the hydrocarbon tails of the soap molecules peg into it, while the negative head is held in water. The grease layer is then dislodged from skin by rubbing or from garment by tumbling and stirring. Each grease globule thus separated is pincushioned by hydrocarbon tails with negative heads outwards in water. The negative globules keep apart by mutual repulsions, and are said to have been emulsified. The emulsified grease globules bearing dirt can be readily washed with water.

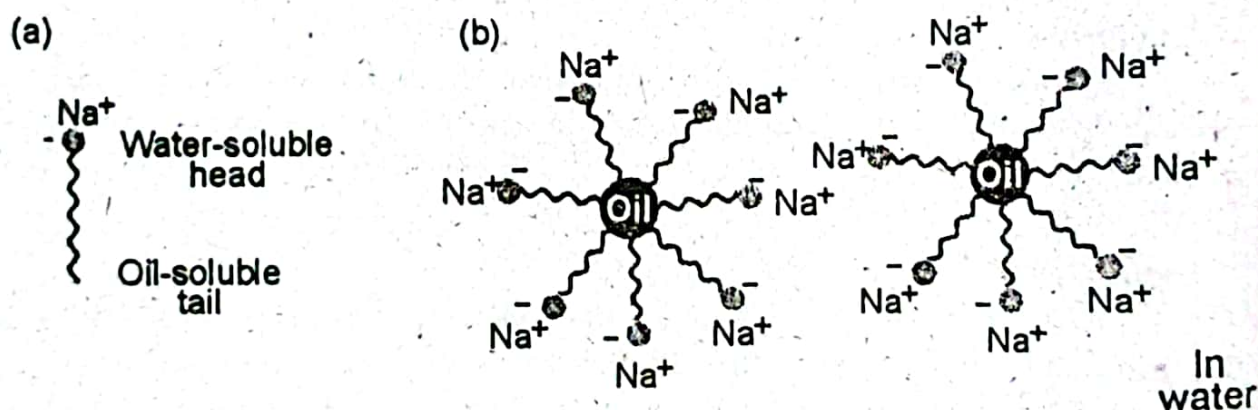


Fig. (a) A soap molecule (b) Emulsified grease globule

Detergents

Synthetic detergents, also known as syndets, or non-soap detergents (NSD) are surface active agents. (surfactants) developed to overcome the drawbacks of soap. They

include all man-made compounds other than soap, which in water solution are surface active and can be used effectively for cleaning.

Synthetic detergents are used mainly for household cleaning and laundry purposes and also in a variety of industries like textile, metal treatment leather, ore flotation, oil drilling, paints, cosmetic, pharmaceuticals, lubricants and agriculture.

The success of synthetic detergents is attributed to their extraordinary properties of stability in hard waters, improved cleaning and availability of raw materials at non-fluctuating rates compared to oils and fats required for soaps. In addition, they can be specially formulated to suit different fabrics and to meet different washing conditions, such as water hardness, volume of water used, water temperature, etc.

Classification

While many thousands of detergents have been patented, only a limited number has become commercially important. These may be classified in various ways; as 'hard' and 'soft' detergents, according to their biodegradable properties (soft detergents are biodegradable whereas hard ones are not), or broadly according to their source, e.g., fatty oil based and mineral oil based.

The electrochemical classification of synthetic detergents is, however, more popular. According to this classification, detergents are grouped as ionic and non-ionic according to whether the surface active portion in the aqueous solution is present and an ion or as a non-ionized molecule. The non-ionic detergent function by a hydrogen bonding mechanism. The ionic detergents are further classified as anionics, cationics, and ampholytes. The latest development in the field of detergent is the enzyme-containing detergent with its special cleaning properties. Some typical detergents are listed in Table 6.1.

Table 6.1. Typical detergents (Anionic and Non-ionics).

Type of detergent	Structure	Reactants
Alkylaryl sulphonate	$\text{RC}_6\text{H}_4\text{SO}_3\text{Na}$	Alkylbenzene. $\text{H}_2\text{SO}_4, \text{NaOH}$
Alkyl sulphate (primary alcohol)	$\text{RCH}_2\text{OSO}_3\text{Na}$	Fatty alcohol, $\text{H}_2\text{SO}_4, \text{NaOH}$
Alkyl sulphate (secondary alcohol)	$\text{R-CH-R'OSO}_3\text{Na}$	Olefines. $\text{H}_2\text{SO}_4, \text{NaOH}$
Fatty amide sulphonate	$\text{RCONHCH}_2\text{CH}_2\text{SO}_3\text{Na}$	Oleic acid, $\text{PCl}_3, \text{H}_2\text{SO}_4$
Fatty ester sulphonate	$\text{RCONHCH}_2\text{SO}_3\text{Na}$	Oleic acid, $\text{PCl}_3, \text{H}_2\text{SO}_4$
Alkyl sulphonate	RSO_3Na	Paraffin hydrocarbons. $\text{SO}_2, \text{Cl}_2, \text{NaOH}$

Alkanolamide	$\text{RaCON} \begin{cases} \text{CH}_2\text{OH} \\ \text{C}_2\text{H}_4\text{OH} \end{cases}$	Fatty acids diethanolamine
Polyoxyethylene alkyl phenol	$\text{RC}_6\text{H}_4\text{O}(\text{CH}_2\text{CH}_2\text{O})_n\text{H}$	Phenol, propylene tetramer ethylene oxide

Anionic Detergents

In anionic detergents, the surface active portion in aqueous solution is present as an anion. The ingredients are mostly derived from fats and petroleum hydrocarbon fractions. This class of detergents include (i) alkylaryl sulphonates (ii) alkyl sulphates (iii) sulphated and sulphonated amides and amines (iv) sulphated and sulphonated esters and ethers (v) alkyl sulphonates.

Alkylaryl Sulphonates

The alkylaryl sulphonates are extensively used for both domestic and industrial purposes because of their low price and remarkable cleaning properties. In textile industry, they are used in wool scouring and wetting. Though their stability and soil suspending power is not as good as those of alkyl sulphates, yet by adding sodium carboxymethyl cellulose and soil-suspending power can be increased.

Up to 1964, the branched alkyl benzene (usually dodecyl or tridecyl – benzene) sulphonates were the ones used in many detergent formulations. These are known as hard detergents (ABS) because of their resistance to breakdown by micro-organisms. They are now being largely replaced by linear (straight chain) alkyl benzene sulphonates (LAS). LAS are preferred because they are more readily biodegradable.

Alkyl sulphates (sulphated alcohols)

The alkyl sulphate series of products includes high foaming detergents and strong wetters as well as effective emulsifiers and dispersants. They are strong competitors to soap because they are stable in acids, alkalies and hard water.

Some of the products are used as lathering and cleaning agents in shampoos and dentifrices while others as detergents and wetting agents for textile processing. They are also employed as emulsifiers and dispersants in emulsion polymerization. Sodium lauryl sulphate prepared from coconut oil represents the bulk of the alkyl sulphates.

Sulphates and sulphonated amides and amines

These substances are water-soluble and give persistent foams. Fatty amides sulphonates are stable to acids and alkalies and are considered better detergents than the fatty alcohol sulphates. They possess very high wetting, cleaning and dispersing properties.

Sulphated and sulphonated esters and ethers

These are emulsifying and wetting agents and also satisfactory detergents. They work well both in acidic and alkaline solutions and are unaffected by calcium or

magnesium ions. In textile industry they are used in kier boiling and scouring operations. A sulpho-ester is one of the most important members of this class. It is prepared from the sodium salt of hydroxymethane sulphonic acid (isethionic acid) and oleyl chloride by condensation. Sulphated monoglycerides of hydrogenated coconut oil have a limited use in ore floatation.

The sulphonated and sulphonated ethers are products of recent development. They are chiefly derived from alkyl polyglycol ethers though fatty alcohol ethers can also be used as bases for sulphonation. These substances compete favourably with alkylaryl sulphonates. However, they are more expensive.

Alkyl Sulphonates

These are prepared by reacting sulphur dioxide and chlorine with petroleum hydrocarbons to give sulphonyl chloride which is subjected to alkaline hydrolysis. They are more stable to acid as compared to alkylaryl sulphonates. However, alkyl sulphonates are not as hard water resistant as alkyl sulphates though their fields of application are similar.

Cationic Detergents

Cationic detergents are those in which the active part of the molecule is positively charged. The products are no doubt, commercially important but have not gained popularity and status equal to the anionic detergents. This is because of the fact that these detergents are more useful as sanitizing and fabric softening agents rather than as cleaning agents.

They have strong lubricating and germicidal properties and are used in surgery as bactericidal agents. They are useful as textile softeners but are poor washing agents as the positively charged high molecular cations have strong tendency to be deposited and retained by many textile and other surfaces which are negatively charged. The quaternary ammonium compounds are the most important members of this class.

Ampholytic Detergents

Ampholytic detergents contain both anionic and cationic groups and as such combine the useful features of both anionic and cationic detergents. Generally, the primary, secondary and tertiary amines, quaternary ammonium compounds, the pyridinium group and imidazoling group act as the cationic group. N-alkyl β -aminopropanoic acid, $R.NH.CH_2.COOH$, hydroxyalkyl methyl taurine $R.CHOH.CH_2.NHCH_2.(CH_2)_2$, SO_3Na and sulphobatins, $R.N(CH_3)_2.(CH_2)_3 SO_3$ may be mentioned as typical members of the ampholytic detergents are finding increasing use in textiles, cosmetics, industrial cleaning and corrosion inhibition.

Non-ionic detergents

These include (i) ethylene oxide condensates (called ethoxylates) prepared from alkyl phenols, fatty alcohols, fatty acids or fatty amines / amides as the base material (ii) polyhydroxy esters (iii) alkanolamides.

Enzyme-Containing Detergents

Addition of enzymes to household detergents, especially those used for laundings is the latest development. Hitherto, detergents had not proved effective on stains caused by milk, tea, coffee, eggs, sweat and blood which even after washing used to leave visible marks knowing the property of different enzymes (protease, amylase etc) to decompose proteins, starch etc. Protease containing detergent compound was introduced in the Netherland in 1963 which required pre-soaking the fabrics before laundering. The enzymes attack the proteins or starches converting them into water soluble compounds which are then washed away by the detergents along with the other dirt. Formulated detergent mixtures containing enzymes have become common both in Europe and USA.

Raw Materials

Unlike in the case of soaps, more than one chemical reaction is involved in the manufacture of synthetic detergents. Hence, raw materials required are diverse and include besides basic raw materials, intermediates, chemical reagents and catalysts. Broadly, synthetic detergents are either fatty oil or mineral oil based.

Both types use one common alkali, either caustic soda or soda ash. They need another very essential reagent, i.e; sulphonic acid. In addition, builders play a very important role in the manufacture of synthetic detergents. Compared to a good soap, which may not contain more than a few percent more than a few percent of builders and good synthetic detergent must be heavily built to the extent of even 80 percent for efficient performance. Infact, without liberal use of some builders not only will the detergents be somewhat ineffective but also expensive. The builders are limited and are made up of alkali silicates, carbonates, sodium sulphate and phosphates (trisodium phosphate, sodium tripolyphosphate and sodium hexametaphosphate). Other additives are optical bleaching agents, chemical bleaching agents and carboxyl methyl cellulose (CMC).

The synthetic detergents industry depends heavily on petrochemicals for its raw material supply of all the known synthetic detergents, the most widely used one is sodium dodecylbenzene sulphonate. Dodecylbenzene is a petrochemical manufactured from propylene and benzene. Propylene gas obtained by cracking of petroleum is polymerized into tetrapropylene which is condensed with benzene to yield dodecylbenzene. Ethylene oxide is another petrochemical used in the manufacture of detergents.

In addition, auxiliary agents, such as whiteners, soil-suspending agents, antistaining agents and perfumes are also used in the manufacture of synthetic detergents.

Manufacture

On account of the large variety of raw materials available, their proper choice and proper designing of equipment with particular regard to the operating conditions of

pressure, temperature and time of reaction are very important in the manufacture of detergents. Both batch and continuous processes are employed.

Batch Process

Concentration (purity and composition) storage and handling characteristics of the chemicals used and correct proportioning and adding of ingredients to the batch are some of the factors which need special attention. The rate at which the ingredients are added to the batch is very important. If they are added too quickly, the time required for chemical reactions or for proper mixing and the total time required to process the batch will be excessive.

Adding some of the ingredients too quickly also increases the danger of overheating the batch, resulting in discoloration of the product. Some of the ingredients are added at intermediate point in the manufacturing process and ingredient should not be added until the preceding process has been completed. Before adding an ingredient to the batch, it is important that both the ingredient and the batch be at the proper temperature. The manufacture of a detergent powder from an alkylate, i.e., dodecylbenzene, is described blow.

Sulphonation

Sulphuric acid, oleium or chlorosulphuric acid are the common sulphonating / sulphating agents. Sulphuric acid is required to be used in large excess (100%), while the other two may be added in 10 percent excess only. In all the cases, spent acid is the by-product which has to be conveniently disposed off. Use of chlorosulphonic acid has the advantage of less evolution of heat, but disposal of the by-product hydrogen chloride poses a problem. Good sulphonation requires careful control of reaction temperature, reaction time, acid strength and ratio of acid to the alkylate.

In a typical sulphonation process, the sulphonating agent is pumped into the sulphonator. The alkylate is then added slowly with vigorous agitation. The sulphonator, which is equipped with an agitator and a heat exchanger, is maintained at about 52°C for 1-2 hours or until the product shows complete solubility. Agitation must be continued throughout the reaction time.

The sulphonation process is complete when all of the alkylate has reacted. After sulphonation, the sulphonator contains a mixture of the sulphonic acid and excess the sulphonator contains a mixture of the sulphonic acid and excess sulphuric acid which gets diluted by the water of reaction. This is the spent acid. If a high percentage of surface active agent in the finished product is desired, the two acids are allowed to separate, the spent acid is drained off and the sulphonic acid is neutralized.

Neutralization

In the neutralization process, a chemical base (or alkali) reacts with the sulphonic and sulphuric acids forming an organic and an inorganic salt respectively. The former is

the actual detergent in the finished product. The function of the inorganic salt and the other chemicals which are added later is to increase the effectiveness of the organic salt as a detergent. Neutralization is carried out in a neutralizer which is a vessel similar in construction to the sulphonator.

In a typical neutralization process, slightly less alkali than is required for complete neutralization is pumped into a reaction vessel. The sulphonic acid mixture is pumped slowly and with stirring into the neutralizing vessel. While the acid mixture is being pumped into the neutralizing vessel, the batch must be cooled to remove the heat of reaction. The cooling rate is adjusted so as to maintain the temperature of the batch at about 40°C. If the temperature exceeds 44°C during neutralization, localized overheating may occur, discolouring the product. After all of the acid mixture has been run into the neutralizing vessels, samples are taken and tested for acidity and alkali is added as required.

Neutralization is complete when the acidity factor of the batch has been reduced to the level called for in the formula. After neutralization the batch is in the form of a slurry which is a solution of the two acid salts in water.

Addition of Builders

A typical household detergent based on ABS or LAS may contain only 20 percent of the active ingredient and the rest is made up of the following components, sodium sulphate 25-30, sodium phosphates (mostly tripolyphosphate) 35 – 40, silicate, CMC, bleaching agents etc. 10 – 15 percent.

The slurry in the neutralizer already contains sodium sulphate. This is taken into account while adding extra sodium sulphate. All the required components are added to the slurry under agitation and the final slurry ready for drying.

Drying

The slurry is spray-dried so that the detergent powder obtained after drying is non-caking, of uniform granular structure and dust free and is of low bulk density. If the slurry is dried by some other method, it is necessary to grind the detergent after drying to obtain a powder. Ground powders are not generally acceptable because they are dusty, non-uniform in size, imperfectly blended with the builders and are slow in dissolving.

The spray dryer comprises a large cylindrical chamber fitted with spray nozzles at the top through which the slurry is pumped into the dryer. The slurry is atomized as it passes through the nozzles and is given a downward velocity. While traveling downward in the dryer rapidly the spray meets a rising stream of hot air. This strips of its moisture with rapidity a designed to cause the small particle of detergent to explode into a hollow bead.

Drying is best accomplished when the hot air flows in opposite direction to the spray. The velocity of the spray as it leaves the nozzles must be great enough to give the solid particles a high enough momentum to pass through the rising hot air and to reach

the bottom of the dryer. If the velocity of the spray is too low, much of the detergent powder will flow out through the top of the dryer with the stream of air and water vapour, making recovery of the powder difficult. If the velocity of the spray is too high, the solids will pass through the hot air so rapidly that they will reach the bottom of the layer without being completely dried and the size of the particles will be smaller. The velocity of the spray is controlled by regulating the pressure in the slurry feed line.

Collecting and storing

The detergent powder passes through the bottom of the spray dryer on to a conveyor which carries it to an air lift. The powder is air lifted to a hopper, dropped to a vibrating screen and then to a storage bin. The dust is collected in a bag filter or other suitable filter.

Continuous Process

The alkylate and oleum taken from storage tanks are fed by means of proportioning pumps into a mixing pump. The outgoing stream from there goes through a heat exchanger (cooler) and is recycled through the mixing pump to bring down the unreacted alkylate content to about 0.2 percent only. The total reaction time is between 5 and 10 minutes.

The reaction mixture is then diluted on its way with measured quantities of water. The diluted solution goes through another heat exchanger to a settler where the sulphonic acid and spent acid separate. The former is continuously fed to a mixing pump along with measured quantities of alkali solution. The mixed stream may be recycled and before discharging it, its temperature is lowered by letting it through a heat exchanger. The emerging steam, i.e., the neutralized sulphonic acid is blended with builder and other additives and spray – dried.

In the continuous sulphonation (sulphating also) using SO_3 instead of oleum an agitated thin layer type reactor is used. The reactor consists of a vertical cylindrical shell (water cooled externally) carrying an internal rotating drum fitted with a series of radial blades which extend to the inner wall of the outer cylinder leaving a very narrow annular space. The alkylate and gaseous mixture of SO_3 and carrier gas (SO_3 , 6%) are fed concurrently into the annular space. Proportioning pumps flow meters and heat exchangers are used for controlling the feeds and temperatures.

The resulting sulphonic acid is led to a digester for ageing to reduce the content of unreacted alkylate. Then follows the stage of "hydration" where in any anhydrides that are formed using sulphonation are hydrolyzed. In the final neutralizing stage, neutralizers acid is fed the alkali (caustic soda, ammonia or ethanalamines). Feed rates, reaction temperature reaction time and pH are automatically controlled. In the SO_3 process no spent acid is produced.

Drying

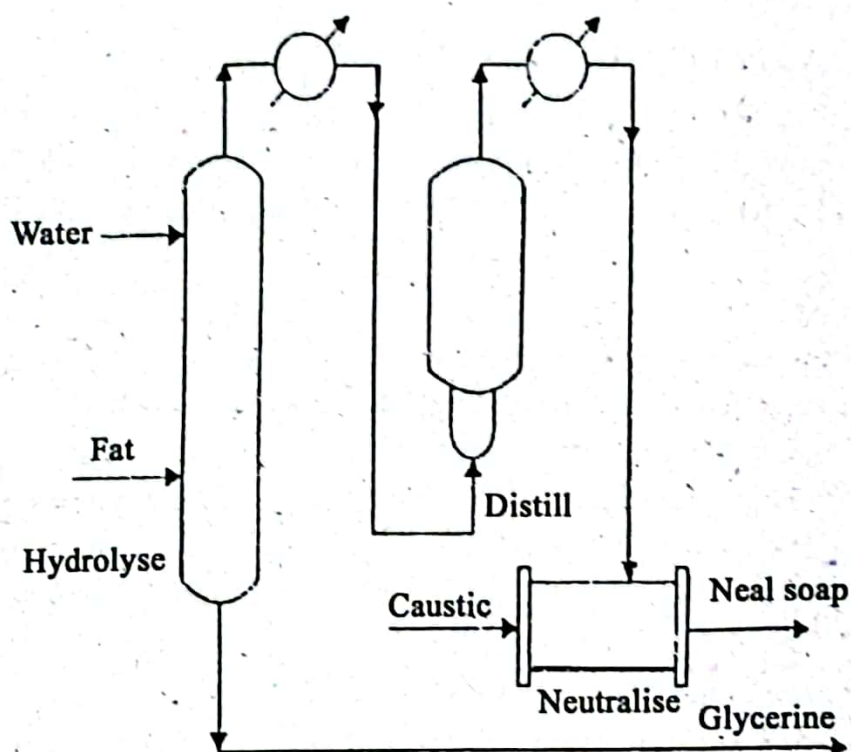
Spray-drying is done just as in batch process to bring down the moisture of the slurry from 30 – 35 percent down to the finished product moisture of 10 percent.

RECOVERY OF GLYCERINE

Glycerine from Saponification Operations

In the making of soap and glycerine by alkaline saponification of fats the glycerine is commonly recovered in solution in the soap lyes. Many of these lyes when removed from the soap pan, contain excess alkali which must be removed before the glycerine can be properly recovered.

Neutralization with acid would be extremely wasteful as it would cause the loss of both the excess alkali and of the equivalent acid. The lye is therefore boiled with fatty matter usually in separate kettles, until substantially all the excess alkali reacts to make more soap, after which it is sent to the glycerine department for treatment and recovery of glycerine. The spent lyes removed directly from the



Basic hydrolyse process for manufacturing soap.

different changes may average round 4 to 5 percent of glycerine but where the soap lye washing is performed counter currently the glycerine in the lye may exceed 10 percent more or less of salt and small amount of soaps that are soluble in the lye. Fig.

Purifying the solution

Soap lyes are given chemical and physical treatments that are calculated to remove as much of the objectionable impurities as possible while clarifying them at the same time. Any of several simple treatments may be employed for this purpose. Most or all of the caustic alkali should have been removed in the soap-boiling department, but the solutions will contain a small fraction of a per cent of sodium bicarbonate and a small amount of soluble soap. The lye may be advantageously brought almost to neutrality with

a small quantity of mineral acid, sulphuric or hydrochloric. It can then be stirred and heated or boiled with some dissolved metallic salt which makes insoluble precipitates with soap and with carbonated alkalies. Salts of this kind which are available and which have been extensively used are ferric sulphate (crude), aluminum sulphate, or ferric chloride in solution. The addition of these reagents is performed under chemical control so that sufficient is employed to remove the last alkalinity and to precipitate any soap present as insoluble soaps, but there will be none of the unused precipitant left. These materials form gelatinous precipitates, either with alkali or with soluble soaps, so that after the precipitate has been removed by filtration in filter presses the lye is clarified as well as nearly neutral.

Evaporation of the solution

The lye is now evaporated, preferably in vacuum evaporators. With the removal of water, the glycerine and salt concentration gradually increases until the salt saturation point is reached, after which salt begins to separate. Further evaporation should be performed in apparatus provided for salt removal. This can be done continuously by the continuous removal of a portion of the settlings from the conical bottom of the evaporator or intermittently by providing the evaporator with a salt drum beneath, with a valve between the evaporator and drum so the salt contents may be removed at intervals. The separated salt may be removed from the liquid portion of the concentrate either by filtration or centrifuging followed by a small amount of washing. The recovered salt is sent back to the soap boiling department to be used again and the concentrated glycerine is boiled down with the removal of more salt, until crude soap lye glycerine is obtained. This will advantageously contain about 80 to 82 percent glycerine, along with about 2 percent non-volatile organic matter; the balance being mostly salt and water.

Treatment of soap lyes with sulphonic acid in the formation of sodium sulphate which find its way into the recovered salt and back to the soap pans. Sodium sulphates not so good for graining soap and for lye washes as sodium chloride. Although the employment of hydrochloric acid and ferric chloride for soap lyes for glycerine recovery is somewhat more expensive than sulphuric acid and sulphate, advantages obtained may outweigh the difference in cost.

In evaporating glycerine lyes there is very little actual volatilization of glycerine in the early stage. In the final stages in bringing the concentrate up to crude glycerine, there is an amount of volatilization of glycerine which though appreciable, need not be large. The actual volatilization of glycerine in evaporation from lye to crude is appreciable less where diminished pressure is employed. This is because the vapour pressure of glycerine relative to that of water is less at the lower temperature. Obviously, the use of diminished pressure at high glycerine concentrations is more necessary for preventing glycerine losses than at lower concentration.

In spite of the fact that very little glycerine need be lost by volatilization in concentrating glycerine lyes to soap lye crude there may be considerable losses in this

operation due to excessive entrainment and poor separation of entrained material from the vapours.

Glycerine Evaporators

Glycerine evaporators are commonly heated by steam in tube calandrias within the lower portion of the evaporators. Diminished pressure is maintained by condensing the vapours with cold water in a barometric column. Non condensable gases are removed from the top of the barometric head either with a wet vacuum pump or an auxiliary steam exhauster.

Many different types of evaporators may be used but whatever the type, it should be such that the heating surfaces will not become coated with salt and adequate salt removal must be provided to reduce entrainment. For this reason it is customary to introduce one or another of various types of "catch-alls" between the evaporator and condenser.

The "catch-alls" should be located well above the liquid level of the evaporator so that any entrained liquor that is separated may flow back into the evaporator by gravity.

An even more important precaution than the introduction of adequate catch-alls is that of avoiding sudden changes in the boiling temperature of the boiling liquor. These may be caused either by irregular introduction of the feed liquor or by unevenness in the pressure within the apparatus. When an unusually large amount of feed is introduced to a boiling evaporator, the feed being more dilute and having a lower boiling point will foam up badly and may even boil over so that a considerable portion of the contents of the evaporator will pass, as foam with great rapidity through the catch-alls to the jet condenser and be lost. A sudden increase in the vacuum may cause a similar disastrous condition. Holding back feed at times will cause the concentration and temperature of the boiling liquor to rise considerably.

This will aggravate the foaming condition when fresh feed is again introduced. Evaporators should therefore be equipped with proper instruments to guide the operators in temperatures, pressure and height of contents at all times. When these are also of the recording type the management can have a check on the operation and find if and where this is at fault so that future operation can be improved.

In some cases, multiple effect evaporators are used in concentrating soap lyes but owing to the fact that the salt which separates is generally greater than the amount of glycerine present and may be irregular in amount, it is doubtful that such evaporators have much if any advantage over single stage evaporators. The various type of crude glycerine whether from aqueous hydrolysis with or without catalyzers or from soap lyes are not pure enough for most uses. In order to obtain good grades of refined glycerines the crude glycerine must be distilled.

Refining of Glycerine

Glycerine boils at 290°C at atmospheric pressure. It is generally stated that glycerine boiling at this temperature is accompanied by decomposition. This is probably not correct or at any rate the decomposition is inappreciable if air is effectively excluded and if local overheating is avoided and if a good excess of steam is present at all times. At a pressure of 10 mm. Hg within the still glycerine boils at about 167°C .

There is such a wide difference in the boiling points of water and glycerine, a difference of about 190°C at atmospheric pressure that separation of water vapour from glycerine has long been practised by fractional condensation. In spite of these differences in physical properties of water and glycerine, Glycerine refiners have been slow in deriving full advantage there from.

Glycerine distilled with open steam at atmospheric pressure is at high temperature and if a concentrated condensate is collected any vapour not condensed necessarily contains a large percentage of glycerine so that only a poor separation is made and any attempt to recover all the glycerine necessitates the condensation of all the direct steam employed along with at least a substantial portion of the glycerine.

It is probable that the greatest improvement in glycerine refinement came from the joint use of direct jet steam and diminished pressure, in conjunction with indirect steam heating of the still contents. Distillation with direct steam at diminished pressure gave greatly improved results both in the quality of distillate obtained and in the ease and thoroughness of separation.

Van Ruymbeke Process

One of the most widely used and one of the most successful methods of distillation is the Van Ruymbeke process. By this process the crude glycerine was heated by indirect steam heat in a still maintained at a diminished pressure amounting to about two to three inches Hg absolute pressure while a considerable amount of direct superheated steam was passed through the still contents.

The mixed glycerine and waste vapour were passed in turn through a series of catchboxes where most of the glycerine was condensed by air cooling of the exterior of the catchboxes, resulting in a number of condensates differing considerably among themselves in amounts of entrained impurities and glycerine concentrations.

The first catchboxes contained appreciable amounts of colour and entrained impurities but with high concentration glycerine and the last catchboxes contained less entrainment but a higher percentage of water and appreciable amounts of volatile fatty acids.

The vapour from the last catchboxes still carried a substantial amount of uncondensed glycerine, the recovery of which was accomplished by the complete condensation of these vapours in surface condensers.

The sweet waters thus obtained contained anywhere from less than 3 percent glycerine to over 5 percent together with the greater part of the objectionable volatile fatty acids which pass over from the still contents, even though these be maintained on the alkaline side. In order to recover the glycerine from sweet waters it is necessary to evaporate off all the water, which is equal to the amount of jet steam used in the still. This necessitates a very decided loss of heat and an appreciable loss of glycerine. The glycerine collected in the last catchboxes is more dilute than desirable for most purposes, so this also must be concentrated by evaporation.

The Modern Process

A system of glycerine refining depending on this principle is now very extensively in use in which substantially all the glycerine volatilized is condensed as distilled analyzing in the neighbourhood of 99.6 percent glycerine without the collection of any sweet waters whatsoever and with no loss of glycerine through volatilization.

All of the glycerine thus distilled is condensed directly into salabale glycerine except as it must be diluted afterwards with distilled water to meet the usual lower commercial standards, and over 80 percent of the product is equal to or better than much of the twice or thrice distilled glycerine. This distilling rate that may be obtained in other stills of the same size and the steam saving is considerable.

The system is quite flexible in as much as it works well over a fairly wide range of operating conditions and the conditions within the system which depend upon one another are largely self-adjusting so that a minimum of manual control is required.

Apparatus

An ordinary still pot, such as one 6 feet in diameter and 6 feet high with dished heads, may be provided with several steam heating coils located within the still in the lower outer portions, the coils being close enough to one another to supply a large amount of heating surface yet far enough apart to give ample room for free circulation of the liquid contents of the still. Five such heating coils may be employed advantageously.

A steam superheater coil may be placed within the other sets of coils for the purpose of superheating the direct jet steam used. The heat employed to superheat the jet steam within the still in this way amounts to only about 2 to 5 percent of the heat required in the still and is less than the heat that is lost by the use of a separator outside superheater.

Located in this position, the superheater coil derives its heat from the still contents just after parting with the glycerine vapour set free as the liquor circulates downwardly over the superheater coil before rising again between the heating coils and does not sensibly lessen the operation of the still. The superheated steam passing downwards through the superheater coil is distributed by a plurality of jets or ejector jets located under the five heating coils so as to discharge the steam with a tangential action.

The tangential discharge of steam from the jets creates a slight circular motion around the still among the coils.

The mixed glycerine and water vapours are passed through an effective entrainment separator that is well insulated to prevent condensation. Then they pass through a special type of surface condenser which is called a boiler-condenser, because of the functions it performs. The vapours passing through tubes are cooled and glycerine is condensed.

The outer sides of the tubes are surrounded by boiling distilled water being generated into steam through recovery of the heat of vapourization of the glycerine. The distilled water supplied to this boiler is the hot condensate from the heating coils employed within the still and when this is released to the lower pressure in the steam-water portion of the boiler-condenser (which may operate at any pressure over a wide range of pressures), an appreciable amount of this water flashes into steam. A portion of the steam generated in the boiler-condenser, from distilled water, is lead to the superheated coil within the still, after passing through a controlling valve and orifice.

It is advantageous to flash the hot, high pressure water from the condensate of the steam heating coils into a chamber maintained at an intermediate pressure at 25 – 30 lbs. gauge and to use the steam thus generated in pre-heating the crude fed to the still.

This type of apparatus will condense somewhat over 80 percent of the distillate as very high grade glycerine testing a little over 99.6 percent pure (and about 0.4 percent water), when the pressure within the still is in the range of 10 – 15 mm Hg. This glycerine require no redistillation to comply with I.P requirements, or to meet most of the special trades specifications.

The mixed glycerine and water vapours passing through the boiler-condenser are now passed upward through a device that might be called a “condenser – concentrator”. It is essentially a shell and tube condenser with the vapour outside the multiple banks of tubes.

Cold water flows through the top tubes and steam from the boiler-condenser flows through the bottom ones, while water at intermediate temperature flows through the others. As the mixed glycerine and water vapour passes upward, some glycerine is condensed out on the hotter tubes, but the water vapour passes on. Part of the water vapour passes out of the top, but part of it along with practically all the glycerine, condenses out on the top tubes. This liquid drips back down over the tubes strikes the next bank, so drips from one row strikes the tubes which are staggered.

The water in the liquid is revapourized before it gets to the bottom tubes. Hence, with proper regulation of temperatures and rates of flow, substantially pure glycerine is produced at the bottom of the condenser and substantially pure water comes off the top. This is really a continuous fractionation by repeated differential condensation.

While it would not be satisfactory for separation of components with a small difference in boiling points it works very satisfactorily for the system glycerine water, there is a great difference in boiling points.

A still of this kind operating on crude glycerine derived from a process like the counter-current hydrolysis of fats, where mineral residues are absent, might operate continuously night and day for over a week with unabated efficiency at a rate of over 3000 lbs, distillate an hour if the feed were available.

Uses of Glycerine

Glycerine is used on a large scale for the manufacture of nitroglycerine, a large part of which is made into dynamite which is probably the most valuable single explosive. Dynamite is used extensively in mining, quarrying and in large construction work. At times, nitroglycerine is extensively employed for military purposes.

Glycerine has a multiplicity of technical uses such as a softening agent, stabilizer of consistency, solvent, dehydrator, lubricant, preservative, and a host of varied physical uses, in making plastics and for many other chemical purposes. It is used extensively in making pharmaceutical preparations and medicinal and is used to a considerable degree in food preparations. Many if not most of these uses will expand if the demands are such as to encourage new uses and more efficient recoveries and if prices can be maintained within a reasonable range.

Methods of safety evaluation of synthetic detergents

Detergents are used almost universally on regular basis. Their safety aspect has been the subject of numerous investigations in the past. The committee responsible for the formulation of this standard felt the need for specifying test methods which can evaluate the safety aspects of synthetic detergents used by the consumers. These test methods have been specified in two parts of this standard. Part 1 of this standard prescribes the test methods for irritant potential of synthetic detergents. These tests are designed to investigate the irritancy potential of detergent products which, during use, involve repeated immersion of hands, forearms and skin contact with the detergent for extended periods of time. The test for skin sensitization is being covered in this second part.

Detergent products which come in contact with skin should be free from skin sensitizing/skin allergy producing components. Such components even when present in minute levels (ppm — part per million concentration) in the formulation, would cause itching, redness, swelling and blister formation on the skin. The onset of such episodes is usually a delayed response from the time of first contact with the product which makes it difficult to identify the incriminating product/ingredient, even by an experienced medical expert.

The detection involves cumbersome procedure of patch testing on skin with all suspect materials on people who are susceptible to such skin allergy. The intractable

nature of this skin condition may cause unnecessary hardships to the consumers. Hence, it is essential that manufacturers of detergent products make sure that their products are free from such hazard, by subjecting their products/ingredients to a predictive skin sensitisation test using a sensitive animal model.

The allergic skin reaction described above is not caused by toxic or irritant properties of the substance (which need to be tested separately) but is due to the increased sensitivity and harmful immunological response to these substances caused by repeated contact. Identification of such allergenic ingredients/ products is possible through the guinea pig maximisation test (GPMT). This standard describes the principle and method of test. Presently, the facilities for these test methods are limited. However, there is scope and imperative need for rapidly expanding and providing these facilities.